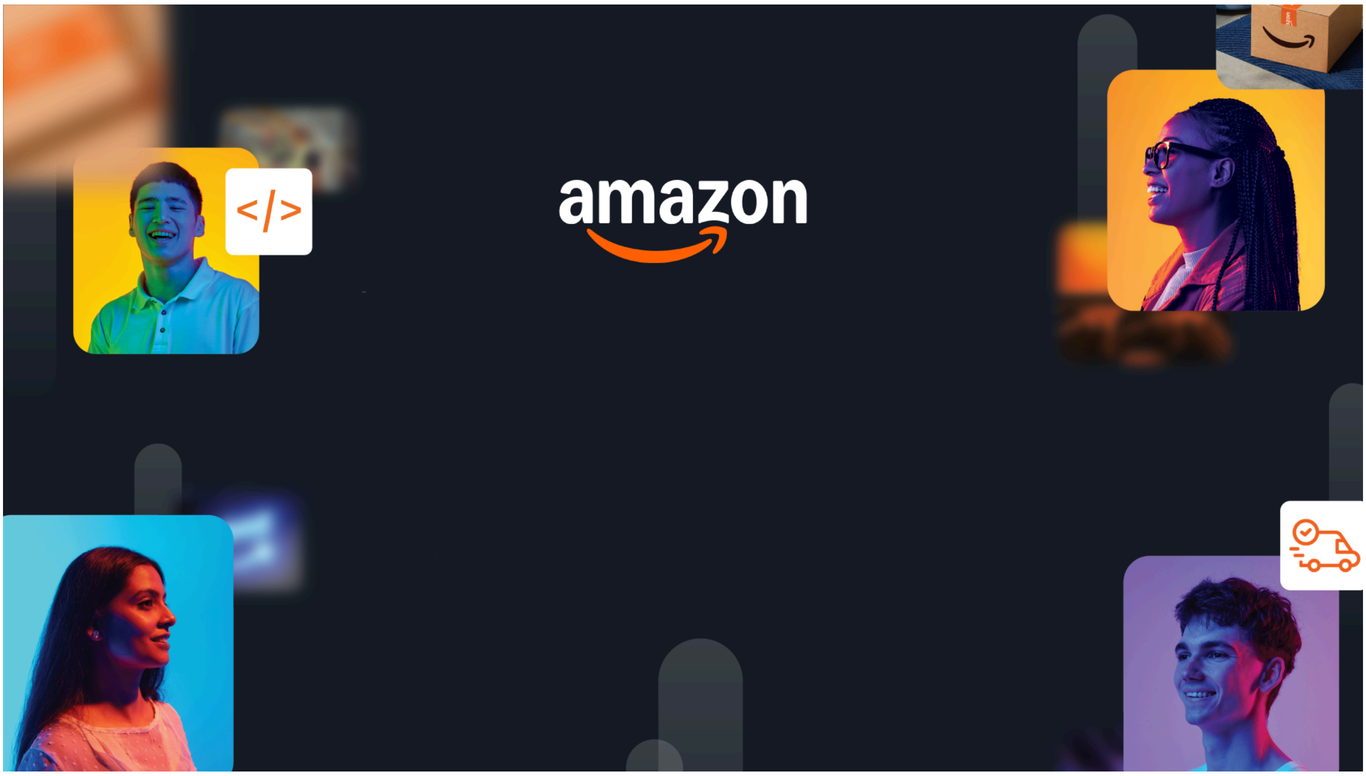

HackOn with Amazon

A Universe of Opportunity

48-Hour Hackathon | Solution Document

Team Name	Codyssey
Hackathon Theme	Amazon Now - Reimagining Urgent Shopping
Date	15/06/2026

Team Members

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1. Problem Statement & Relevance

The Problem

- Quick commerce made delivery fast, building the cart is still slow.
- India has 290–300 million online shoppers (Bain & Company, 2025), and they spend 5–7 minutes per order just searching items, comparing variants, and finding substitutes when something is out of stock (RedSeer, 2025).
- The delivery takes 10 minutes. The decision takes just as long — and no platform solves it. Every app still forces you to pick items one by one. “NowCart” fixes the cart-building problem: give it a need, get back a ready-to-checkout cart.

Why It Matters

- **Who:** 300M+ Indian online shoppers (Bain & Company, 2025) spend 5–7 minutes per session assembling carts for meals and restocks they already have in mind — across Blinkit, Zepto, Instamart, and Amazon Fresh.
- **How large:** This "deciding time" sits on top of a $7B+ quick-commerce market in India (Mordor Intelligence, 2025) that perfected 10-minute delivery but left the cart-building bottleneck completely unsolved.
- **Cost of inaction:** Around 70% of online shopping carts are abandoned before checkout (Dynamic Yield, 2025) — that's 7 out of 10 carts where the user gave up. For platforms, every abandoned cart loses sales revenue. Every out-of-stock dead-end erodes trust, and every frustrating session pushes users to switch apps permanently.

Theme Alignment

Theme connection — NowCart addresses all three opportunity areas with one coherent product:

- **Frictionless Shopping —** out-of-stock items are substituted automatically and transparently. The user never hits a dead end or gets stuck comparing six variants of the same product.
- **Shopping by Intent —** snap a dish photo or paste a recipe/YouTube link — get a cart. For new users, assistive prediction uses age and gender from signup to build a starter cart.
- **Predictive & Confident —** returning users get a predicted restock cart before they ask. Every pick shows a confidence score and a one-line reason.
- **Integration-ready —** voice via Alexa, image via Amazon Photos, engine exposed as an API. Plugs into existing ecosystems.

Unique angle — Every platform works from product → requirement. NowCart **reverses** it: **requirement → product.** Tell us what you want to make, we assemble what you need to buy.

Example: Don't know what goes into penne pasta? Just say "penne pasta for 2" — “NowCart” builds the cart.

What Makes This Novel

Core novelty — Informed Choice, Not Forced Choice. Instead of showing six variants of ghee and leaving the user overwhelmed, we rank results by relevance and surface the highest-confidence picks with a NowCart Verified badge (most ordered in category + highest rated). The user still picks but now with clarity, not chaos. No bias toward any brand, no hidden selection by AI.

What's architecturally new — A **recipe link**, **dish photo**, **budget**, or **voice** command or **subscription** — all converge into a single reasoning framework. Each step builds on the previous: match → semantic re-rank → score. If any step fails, the pipeline gracefully skips it and proceeds with what it has so no single point of failure breaks the cart. No grocery platform has a reasoning framework instead of that they have search indexes.

2. Customer & Solution

Target Customer

Who we're building for — three distinct user groups**:**

User Group	Their Problem	NowCart Feature That Solves It
Never-in-the-kitchen people	Wants to cook something they saw online but doesn't know the recipe or ingredients	**Show** + **Share** — snap a dish photo or paste a YouTube/recipe link, get the full ingredient cart
Elderly / voice-first users	Can't navigate text-heavy apps, struggle with small buttons and typing	**Speak** — just say "dal chawal for 2" and get a cart
General household shopper	Knows what to cook but spends 5–7 min searching each item, comparing brands, handling out-of-stock	**Constrain** + **Subscribe** — set a budget to fit the meal, or set recurring items and let the system predict restocks from past orders

What they all need — Not a better search box, but an interface that takes a simple need and hands back a ready cart. One missing item never breaks the flow, it's swapped with the next best match transparently.

How We Solve It

"NowCart" turns a plain need into a checkout-ready cart. Five ways:

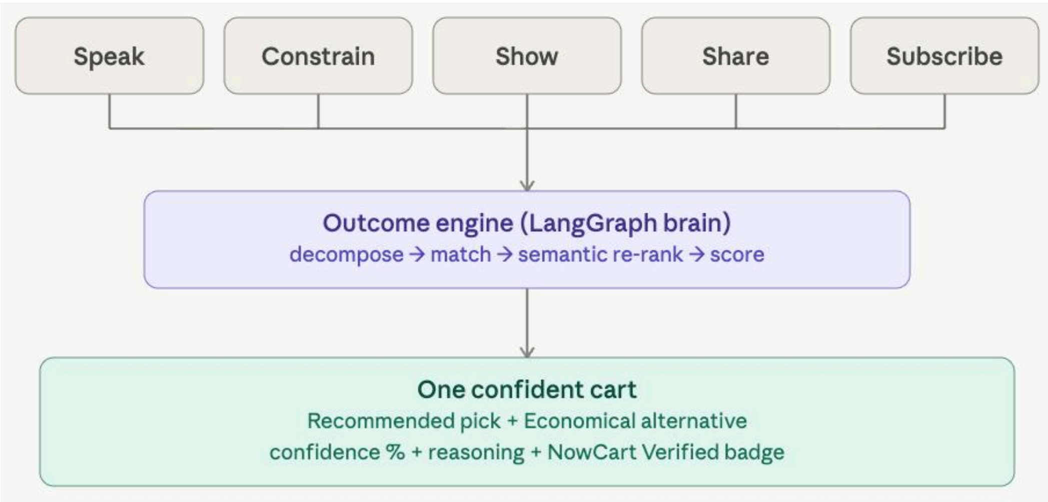
1. **Multiple front doors, one engine —** every input feeds the same framework:

Front Door	What the user does	What NowCart does
Show	Snaps a photo of a dish	Extracts ingredients, builds the cart
Share	Pastes a recipe link or YouTube URL	Parses the recipe, builds the cart
Speak	Says "biryani for 4"	Understands the intent, builds the cart
Constrain	Sets a budget (e.g. ₹500)	Works backwards to fit the budget
Subscribe	Sets recurring items (e.g. milk daily) or does nothing	Auto-adds recurring items and predicts restocks from purchase patterns — before you ask

2. **Trusted picks, user decides —** instead of 20 lookalikes, results are ranked by relevance with trusted seller badges (like "Nowcart Trusted") so users can pick with clarity. A cheaper economical option is always shown alongside. The user always makes the final choice.

3. **Resilient and pantry-aware —** Subscribe builds a restock cart from your purchase patterns and lets you set recurring cycles per product shifting from reactive to anticipatory.

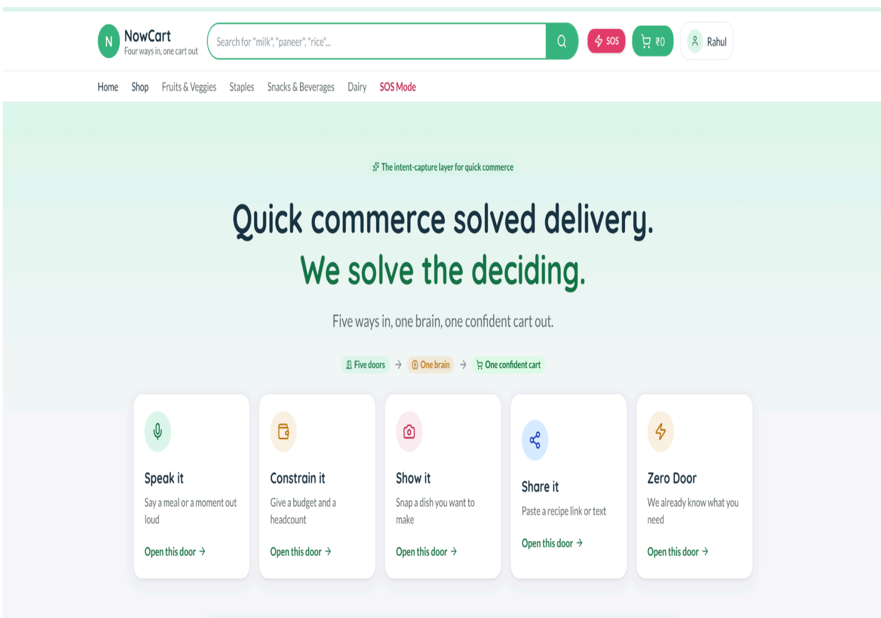
User Workflow

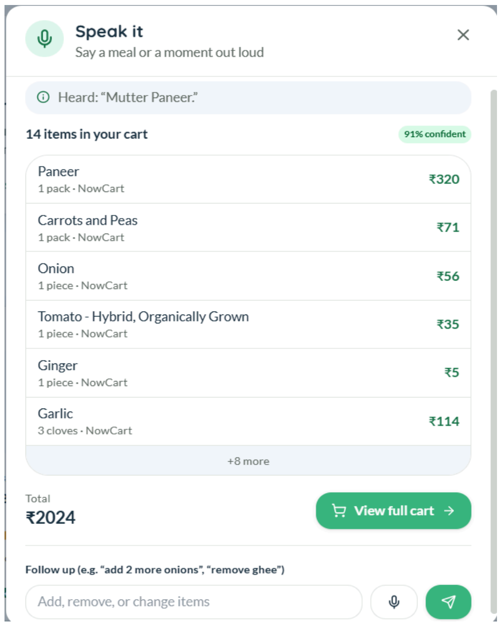


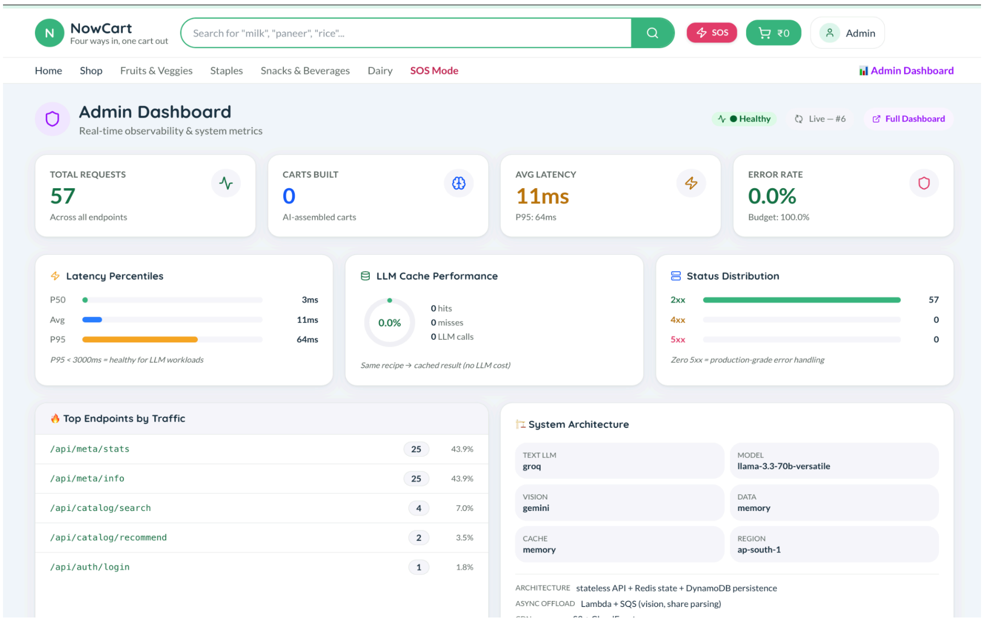
3-step flow in words:

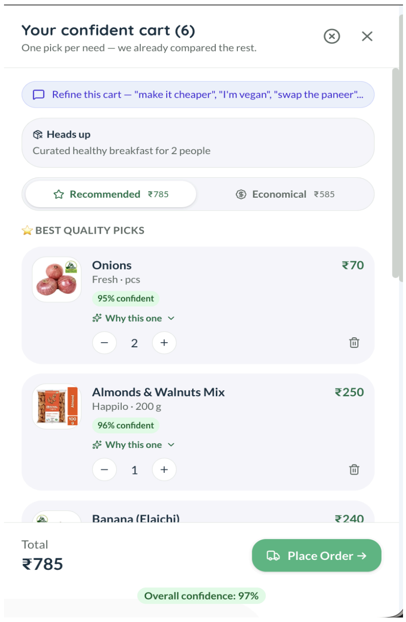
1. **Express:** open any front door: speak a meal, set a budget, snap a photo, paste a link, or just let it predict.
2. **Engine:** the pipeline decomposes the intent, matches it against 9,534 products, optimizes the picks, resolves out-of-stock items, and scores its confidence.
3. **Confirm:** one cart appears with confidence scores, a reasoning trail, and transparent substitutions — then the user refines it conversationally ("make it cheaper," "I'm vegan") or checks out in two taps.

Working Prototype







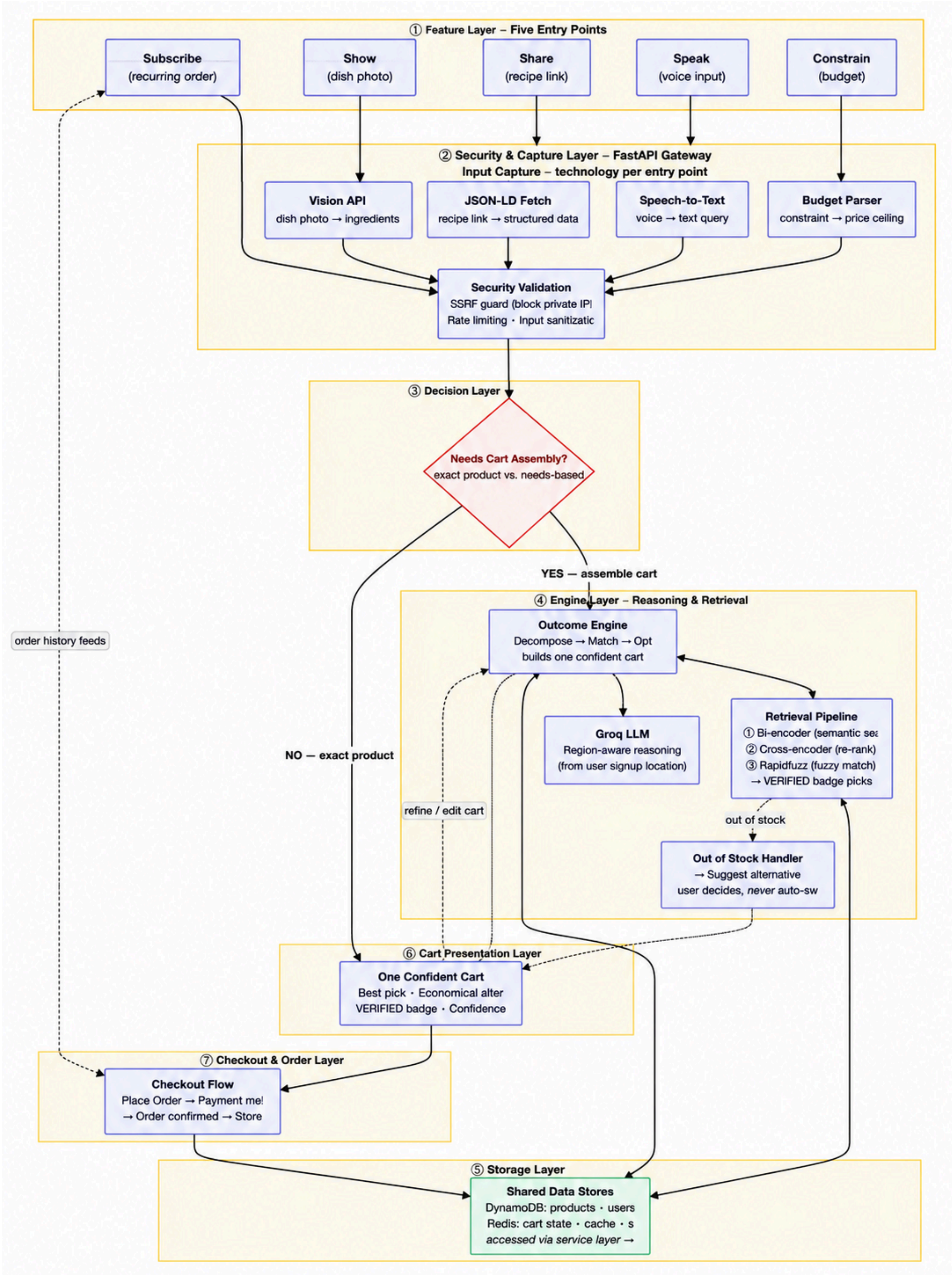


Demo: [Demo Video / Live App / Source Code]

Deployed on AWS and fully functional. For the complete experience (pantry awareness, predictions, order history), sign in as rahul@gmail.com — any password works.

3. Tech Architecture & Scaling

Architecture



Tech Stack

Layer	Technology	Why
Frontend	React 19 + Vite + TailwindCSS	1. React 19 + Vite keeps all five front-door panels instant and lag-free. 2. TailwindCSS ensures fast load times even on mid-range phones with patchy connections.
Backend	FastAPI + Pydantic 2 (fully async)	1. FastAPI handles thousands of simultaneous cart requests without anyone waiting. 2. Pydantic ensures all data between the app and server is validated automatically before it reaches the user.
AI / ML	LangGraph + Groq (Llama 3.3 70B) / Gemini 2.0 Flash / Bedrock (Claude 3 Haiku) + pure-NumPy TF-IDF & rapidfuzz	1. LangGraph drives the step-by-step reasoning that turns any user intent into a confident cart, with swappable AI models underneath. 2. Fuzzy search that catches typos and Hindi terms like "malai" without any heavy downloads.
Database	DynamoDB (PAY_PER_REQUEST)	1. DynamoDB scales automatically to any number of users and orders with zero manual effort, and stores order history that directly powers “Subscribe” restock predictions for returning users.
Infra	EC2 + Nginx + S3 + CloudFront + Redis (Lambda + SQS designed)	1. CloudFront delivers the app fast from anywhere in the world. 2. Redis makes repeat queries feel instant, and Lambda + SQS offloads heavy AI tasks so the app stays responsive even at peak traffic.

Key Algorithms & Complexity

- **LangGraph Multi-Agent Pipeline** — A 10-node reasoning graph with shared typed state and a self-correcting replan loop (capped at 2 passes). Each node builds on the last, so the logic is composable and debuggable instead of one brittle mega-prompt. Complexity: linear in needs × catalog size.
- **Hybrid Retrieval (Bi-encoder + Cross-encoder + Fuzzy)** —A bi-encoder (all-MiniLM-L6-v2, 80MB) encodes query + products into vectors and finds the top-20 by semantic meaning . A cross-encoder (ms-marco-MiniLM-L-6-v2, 80MB) re-ranks those 20 by pairwise comparison . Rapidfuzz handles typos as a fallback .Example: Catches "cottage cheese → paneer" (semantic), "malai → cream" (cross-language), and "tomatoe → tomato" (typo). Complexity: linear embedding pass, then constant re-rank on shortlist.
- **Confidence-Sorted Budget Fill** — Items are ranked by confidence and kept greedily until the budget runs out, so the most trustworthy picks always survive the cut. Complexity: one sort plus one pass.
- **Subscribe (Predictive Restock + Recurring Schedules)** — Predicted restock measures inter-purchase intervals, projects depletion via coefficient of variation scoring, and ranks certainty by pattern regularity — pure statistics, no training, no cold start. Recurring schedules resolve on a fixed per-product cycle at cart-load time. Complexity: linear in products × order history for prediction; O(1) lookup for schedules.
- **LLM Response Caching** — Each prompt is hashed to a 1-hour Redis entry, so 100 identical queries cost 1 model call and 99 sub-millisecond cache hits — cutting both spend and ~800 ms of latency per repeat.

Scaling Strategy

- **Stateless API:** Cart state lives in Redis, not in the server process, so any instance can serve any request. 100x: an Auto Scaling Group behind a load balancer adds instances on demand. 1000x: multi-region groups with latency-based routing send users to their nearest region.
- **Database:** no capacity planning. DynamoDB on-demand auto-scales reads and writes natively, with no sharding or provisioning. 1000x: Global Tables replicate across regions for fast local reads worldwide.
- **LLM offloading (scaling path):** Slow AI calls (vision ~3s, recipe parsing ~2s) move to Lambda via SQS, auto-scaling to 1000 parallel executions. This decouples API response time from AI processing time entirely — the API stays fast no matter how heavy the model work gets.
- **Multi-layer caching:** In-memory hot catalog data, Redis for carts and LLM responses (hashed, 1-hour TTL), and DynamoDB DAX for catalog reads at scale. Common intents like "biryani for four" become pure cache hits — instant and free.
- **Frontend:** already global. S3 + CloudFront serves the app from edge locations worldwide today, with no change needed to handle millions of users.
- **Provider abstraction:** Switching from Groq to Amazon Bedrock is a single environment variable. At scale, Bedrock runs inside our AWS network with managed throughput and no third-party rate limits.

4. Future Vision

Where This Goes

NowCart becomes the universal "need → done" layer on top of any fulfillment network. The engine is category-agnostic — it works anywhere a person has a goal and needs products assembled, not just groceries.

Roadmap

Horizon	Milestone	Impact
0-3 mo	Grocery intent-capture live in one Indian metro — voice, budget, photo, and share all functional	1K active users, 3x faster cart assembly vs. manual, under 15% cart abandonment
3-6 mo	Pharmacy catalog integration (OTC products), multi-language voice (Hindi, Tamil, Bengali), predictive restock at scale	10K users, language barrier removed for 500M+ non-English speakers
6-12 mo	B2B restocking (restaurants, offices), event and travel kits, white-label API integration with grocery platforms	20k+ users, licensing revenue per cart built, horizontal "need → done" layer proven

Multi-Segment Expansion

NowCart's engine is domain-agnostic: it applies to any market where people know the outcome they want but not the exact products to buy. The expansion path moves from consumer essentials to high-frequency commercial use:

- **Grocery (today)** → "Biryani for 4" becomes a full ingredient cart in seconds — proving the core thesis.
- **Pharmacy & Health** → "Child has a fever" → OTC medicine, hydration, and a comfort kit. The highest-urgency, highest-trust market — same pipeline, health catalog.
- **Creator & Social Commerce** → any cooking video, reel, or blog becomes a shoppable cart. We already turn a YouTube link into ingredients today; the path is partnering with creators and platforms so recipe content is one-tap buyable. The content already exists — we make it purchasable.

- **B2B Restocking** → "Restock my restaurant for the week" or "office pantry for 50" → bulk carts with quantity scaling, budget limits, and recurring auto-orders. Higher order values, predictable recurring revenue.
- **Travel & Event Kits** → "Camping trip for 6" or "party for 20 kids" → a single cart spanning food, supplies, and gear across categories.

The Path: consumer (grocery) → urgent care (pharmacy) → content-driven (creator commerce) → commercial (B2B) → cross-category (events). Each step reuses the identical engine — adding a vertical is a catalog and prompt change, not an architecture rebuild. Any industry where humans have a goal but face product overload is addressable.

Value Impact

- **Users reached.** Approx 1.4B+ (Statista, 2023) online grocery shoppers globally (71M+ in India (ResearchGate, 2025)), with multi-segment expansion adding 200M+ across pharmacy, B2B, and travel. Critically, voice-first and photo-first input unlocks an estimated 100M+ users excluded by text-only interfaces — the elderly, visually impaired, and non-English speakers.
- **Time saved.** ~10 minutes per session × 3 sessions/week ≈ 26 hours returned per user per year. At 1M users, that's 26M hours of human time given back annually — an entire friction layer removed from daily life.
- **Efficiency gains.** Confident carts with pre-resolved out-of-stock items directly attack cart abandonment, an industry-wide ~70% problem in online retail (Baymard Institute, 2025). LLM response caching removes up to 99% of redundant model calls at scale, so cost stays flat as repeat queries grow.
- **Revenue potential.** A white-label "cart-built" pricing model: at a few cents per cart and 10M carts/month, that's a six-figure monthly licensing revenue stream — on top of faster carts driving higher order frequency and platform Gross Merchandise Value for partners.

Links: *GitHub* / *Demo Video* / *Live App*